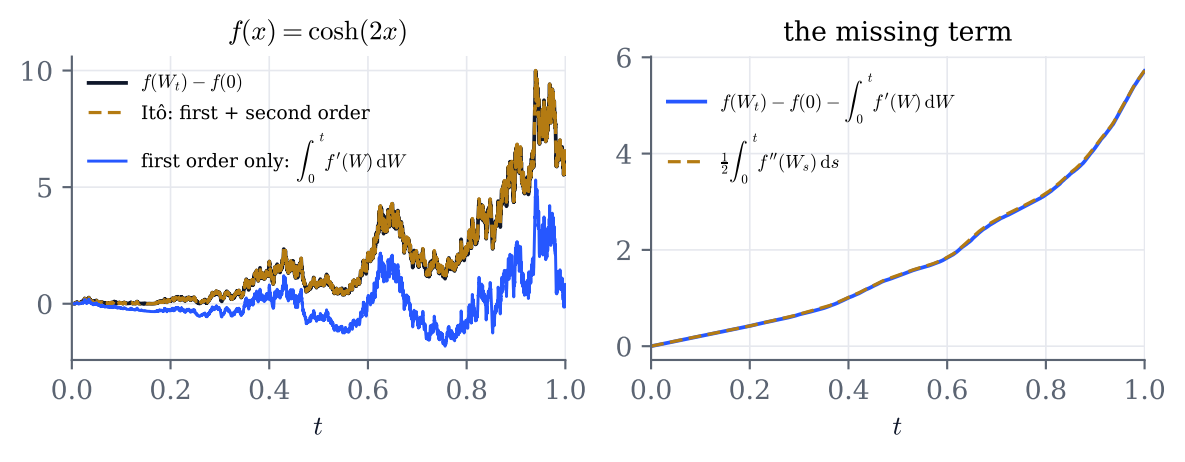

$f(x) = \cosh(2x)$
$f(W_t) - f(0)$
Itô: first + second order
first order only: $\int_0^t f'(W)\,\mathrm{d}W$
10
5
0
0.0
0.2
0.4
0.6
0.8
1.0
t
the missing term
$f(W_t) - f(0) - \int_0^t f'(W)\,\mathrm{d}W$
$\frac{1}{2}\int_0^t f''(W_s)\,\mathrm{d}s$
6
4
2
0
0.0
0.2
0.4
0.6
0.8
1.0
t